

Electrical Engineering Student

Nir Cohen

Profile

Fourth-year Electrical Engineering student at Tel Aviv University specializing in signal processing and machine learning. Proficient in Python, C++, and MATLAB, with hands-on experience building algorithmic pipelines for audio processing, time-series analysis, and complex system modeling. Curious, detail-oriented, and motivated by leveraging strong mathematical foundations to extract insights and build robust predictive models. Collaborative and always looking to learn from teammates and grow as an engineer.

Education

Tel Aviv University — B.Sc. Electrical Engineering | GPA: 83 (03/2023 – 2027)

The Open University — Academic Pathway Program | GPA: 85 (09/2020 – 03/2023)

Projects

Music Source Separation via NMF (Python, Signal Processing)

[GitHub](#)

- Implemented Non-negative Matrix Factorization (NMF) from scratch to decompose audio signals into individual instrumental components.
- Developed an automated pipeline for Spectrogram pre-processing, utilizing STFT (Short-Time Fourier Transform) and specialized activation/basis matrix initialization.

Real-Time Piano Note Recognition System (C, Arduino, Signal Processing)

- Designed and implemented note recognition using harmonic frequency analysis, leveraging hardware-level features for real-time performance.
- Employed a dynamic scoring algorithm to determine the most likely note based on multiple frequency harmonics.
- Achieved top grade (100) in Hardware Lab.

PID Controller Implementation (C++)

- Implemented PID controller in C++ for precise control of an inverted pendulum car.

Academic Highlights

- Data Structures & Algorithms – 98
- Signals & Systems – 86
- Linear Algebra – 94
- Digital Logic – 86

Experience

Lab Technician | IDF Air Force | Hatzor Airbase

(04/2020 - 12/22)

- Maintained and calibrated Scanning Electron Microscope and Color Spectrometry Equipment for propulsion system testing. Performed laboratory testing
- Designed an optimization fixture, reducing test times and waste; Received an **award** for innovation

Skills

Programming: Python (4+ yrs), C, C++, MATLAB

ML & Signal Processing: NumPy, SciPy, PyTorch

Tools & Platforms: Linux, Raspberry Pi, Arduino

Languages: Hebrew (Native), English (Fluent)